



جامعة الكويت
KUWAIT UNIVERSITY

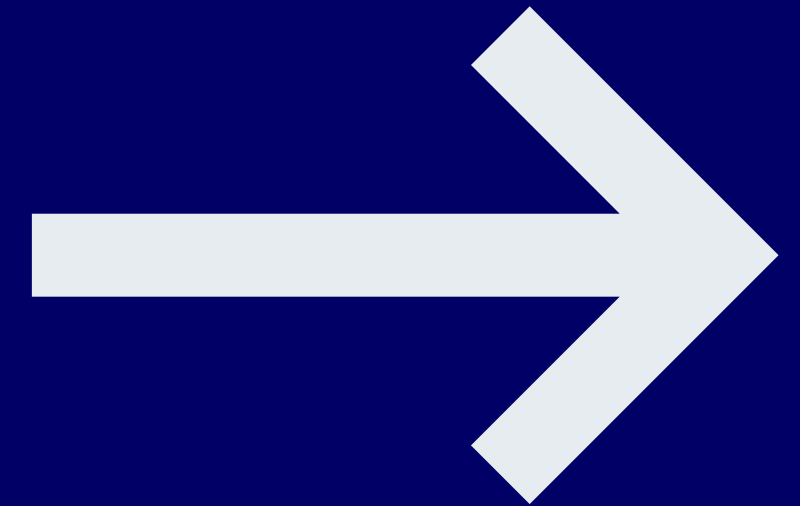
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Pulse of the Campus: Real-Time Health Tracking with Bangle.js 2

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Introduction

In smart campus environments, wearable IoT devices like the Bangle.js 2 smartwatch enhance health, safety, and overall user experience. The proposed system enables real-time monitoring of steps, heart rate, and motion, streaming data directly to a browser dashboard through the Web Bluetooth API without requiring a native app.

The platform tracks both individual and group activities, integrating gamified features such as step competitions to promote wellness throughout the Kuwait University Campus. It supports real-time analytics, activity heatmaps, and predictive health insights, with future potential for machine learning-based behavioral pattern analysis and fatigue detection.

Objectives

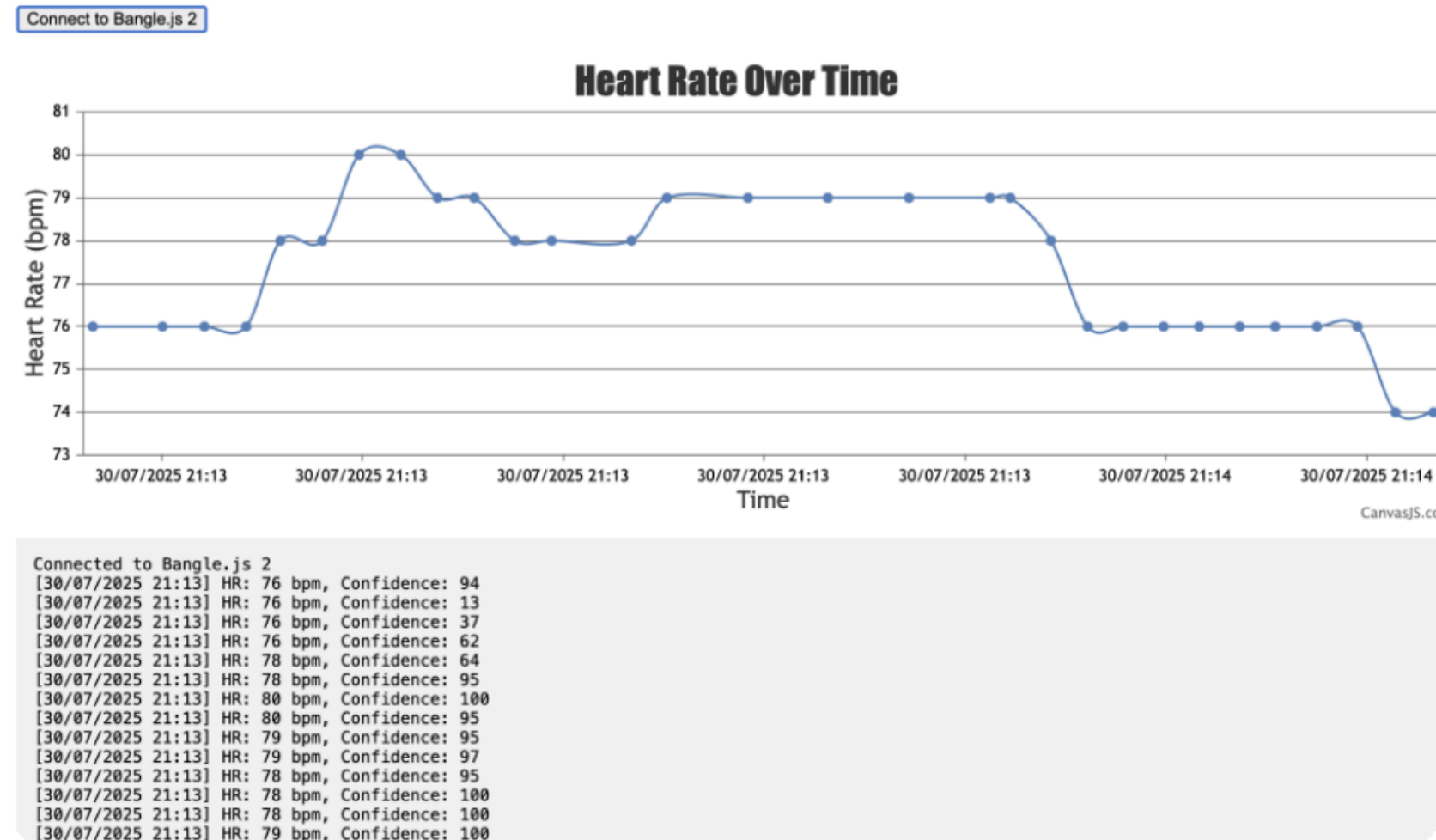
- To develop a proof-of-concept for a real-time health and activity monitoring system using the Bangle.js 2 smartwatch.
- To demonstrate the integration of wearable IoT data with a browser-based dashboard for continuous visualization and analysis.
- To assess the feasibility of deploying such a system for campus-wide wellness tracking and data-driven planning within Kuwait University.

Implementation



Bangle.js 2 Continuous Data Capture

Vital values such as user's heart rate, ambient temperature, and other environmental data

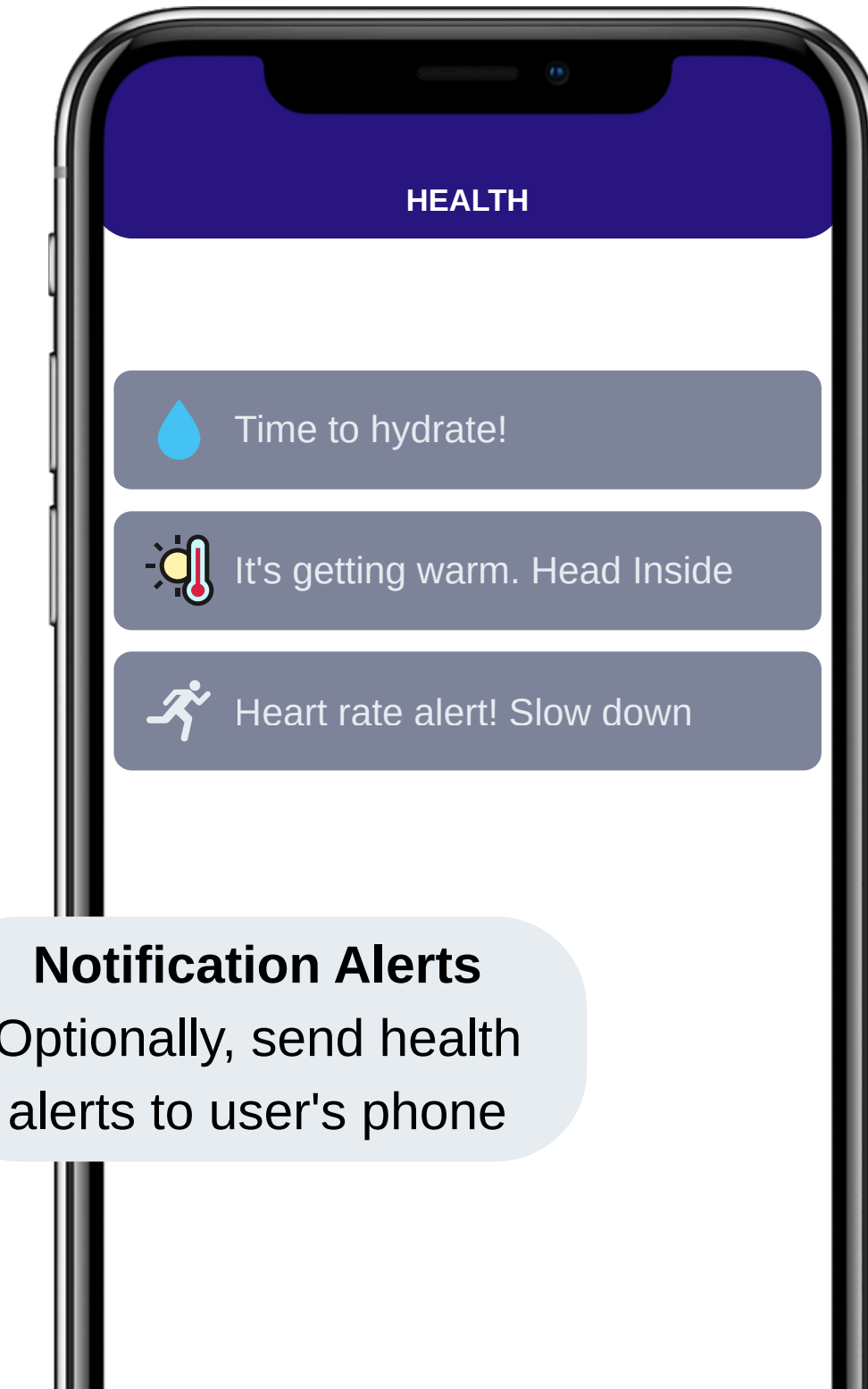


Browser-based Analysis

Analyse real-time physiological and environmental data

Process Data & Input to Model

Thorough data processing to pass input to the ML model for predictive health alerts



Notification Alerts

Optionally, send health alerts to user's phone

Future Scope

- Predict heat exhaustion and fatigue using real-time data and machine learning.
- Integrate with campus GIS for location-based activity and health mapping.
- Provide accurate, AI-driven health alerts and personalized wellness insights.
- Enable data-driven campus planning based on collective activity trends.
- Support advanced analytics for early detection of health risks.

Conclusion & Key Takeaways

- The Bangle.js 2-based system proves that real-time health and activity tracking on campus is both feasible and valuable.
- Using the Web Bluetooth API, data can be visualized directly in the browser—eliminating the **need** for a native app.
- Gamified elements such as step competitions foster engagement and promote a culture of wellness across the university.
- The platform lays the groundwork for AI-driven insights, predictive health alerts, and data-informed campus planning.
- Ultimately, this work shows how IoT and AI can turn a smart campus into a living health ecosystem—one that is healthier, safer, and more connected.



**Thank
you!**

**Any
questions?**

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